

Functional Analysis

Selected Topics on Mathematical Physics | Autumn 2026

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I Prerequisites

Students are expected to be familiar with the following topics:

- Complex analysis
- Linear Algebra

II Course Structure

The course meets once per week for lectures. In addition, a weekly tutoring session is held to support work on the problem sheets; attendance is strongly encouraged, as the sheets form an integral part of the course.

	Format	Duration
Regular session	Theory and derivations	80 minutes
Tutoring session	Problem sheet workshop	90 minutes

Problem sheets should be attempted *before* the tutoring session. Hints and solutions are discussed in the tutoring session and collected shortly thereafter.

III Course Description

This course develops the modern foundations of analysis, beginning by laying the logical and topological groundwork — set theory, countability, and the rudiments of metric and topological spaces — before developing a rigorous theory of measure and integration. Abstract measure spaces are constructed from first principles, and the Lebesgue integral is built up carefully, with the Monotone Convergence Theorem, Fatou’s Lemma, and the Dominated Convergence Theorem as its cornerstones. Differentiation is then revisited in this modern setting via the Hardy-Littlewood maximal function and the Lebesgue Differentiation Theorem. The middle part of the course broadens the analytic landscape in two directions. Distribution theory introduces generalised functions — including the Dirac delta distribution and the generalised derivative — along with Fourier transforms and the Schwartz space, providing tools essential to modern PDE theory. In parallel, the abstract framework of Banach spaces is developed: L^p spaces, Hölder and Minkowski inequalities, the Riesz-Fischer theorem, and the role of completeness. The course culminates in the theory of Hilbert spaces and their operators. After establishing the geometry of inner product spaces — Cauchy-Schwarz, orthogonality, and the centrality of L^2 — we conclude with the study of bounded linear operators on Hilbert spaces, including adjoints, the spectrum, compact operators, the Fredholm Alternative, and the Spectral Theorem.

IV Contents

1. **Countable Sets** — Review of set-theoretic foundations; finite, countable, and uncountable sets; Cantor's theorem and diagonalization arguments.
2. **A Glimpse into Topology** — Topological spaces; metric topology; basis for a topology; convergent sequences; Cauchy sequences.
3. **Measure Spaces** — The problem of measure; measurable spaces; Borel sets; abstract measure spaces $(X, \mathcal{M}, \mu)$; properties of measures; probability spaces; measurable functions; the extended real numbers; the Lebesgue σ -algebra.
4. **Abstract Integration** — Integration of measurable functions; construction of the Lebesgue measure; comparison with the Riemann integral; integration over Lebesgue-measurable subsets of $\mathbb{R}$; monotone Convergence Theorem; Fatou's Lemma; Lebesgue's dominated convergence theorem; interchange of limits.
5. **Distribution Theory** — The space $\mathcal{C}^\infty$ of test functions; distributions (generalised functions); regular distributions; Dirac's delta distribution; the generalised derivative; Fourier transforms; the Schwartz space $\mathcal{S}$.
6. **Banach Spaces** — Metric spaces; review of linear spaces and linear functionals; linear functionals; $\mathcal{L}^p$ and L^p spaces for $1 \leq p \leq \infty$; Hölder and Minkowski inequalities; completeness and the Riesz-Fischer theorem; pre-Hilbert spaces; complete inner product spaces; Cauchy-Schwarz inequality; the space L^2 of square-Lebesgue-integrable functions.
7. **Hilbert Spaces** — $\mathcal{L}^p$ and L^p as linear spaces for $1 \leq p \leq \infty$; Hölder and Minkowski inequalities; completeness and the Riesz-Fischer theorem; pre-Hilbert spaces; complete inner product spaces; Cauchy-Schwarz inequality; the space L^2 of square-Lebesgue-integrable functions.
8. **Linear Maps on Hilbert Spaces** — Operators on Hilbert spaces; adjoint and inverse operators; bounded operators; spectrum of an operator; compact operators; Fredholm alternative; spectral theorem.

This syllabus is subject to change. Any modifications will be announced in class.

V References

The star rating reflects the mathematical sophistication of each text: (★) accessible with the course prerequisites; (★★) requires additional mathematical maturity; (★★★) intended for readers revisiting the subject — not recommended if this is your first encounter with the material.

- [1] Axler, S. *Measure, Integration & Real Analysis*. 1st ed. Springer, 2020.
[Primary (★)]
- [2] Le Gall, J.-F. *Measure Theory, Probability, and Stochastic Processes*. 1st ed. Springer, 2022.
[Primary (★★)]
- [3] Szekeres, P. *A Course in Modern Mathematical Physics*. Cambridge University Press, 2004.
[Supplementary (★★★)]
- [4] Bartle, R. G.; Sherbert, D. R. *Introduction to Real Analysis*. 4th ed. Wiley, 2011.
[Background reference (★)]